

SWS3026 Visual Computing

Tutorial 3: Digit Recognition

The MNIST database (Modified National Institute of Standards and Technology database) is a large database of handwritten digits that is commonly used for training various image processing systems.

Tasks:

1. Familiarize yourself with the MNIST dataset: [MNIST handwritten digit database, Yann LeCun, Corinna Cortes and Chris Burges](#)
2. Familiarize yourself with sklearn package: <https://scikit-learn.org/stable>
3. Study SVM classifiers: <https://scikit-learn.org/stable/modules/svm.html>
4. Study k-Nearest Neighbours classifiers: <https://scikit-learn.org/stable/modules/neighbors.html>
5. Study RandomForest classifiers: <https://scikit-learn.org/stable/modules/ensemble.html#forests-of-randomized-trees>
6. Study Naïve Bayes classifiers: https://scikit-learn.org/stable/modules/naive_bayes.html

Programming exercises:

Q1. Use the `fetch_openml` function found in `sklearn.datasets` to load the **mnist_784** dataset into python. That is, `mnist = fetch_openml('mnist_784', parser='auto')`

This will load `X` (`mnist.data`) and `y` (`mnist.target`) variables for you.

- Print the dimensions of the variables returned by the function, ie. `mnist.data.shape`, `mnist.target.shape`
- Write a python script to find how many distinct values are present in `y`?
- Select one sample from `X` for each distinct `y` value.
- Resize each sample to represent the 28x28 pixel image.
- Display all the selected images in one diagram using subplots in `matplotlib`. The following code gives you an example of how to do this,

```
fig = plt.figure()
for i in range(1, 11):
    fig.add_subplot(2, 5, i)
    plt.imshow(images[i])
plt.show()
```

Q2. Use sklearn to train a digit classifier.

- Split the X and y into a training set and testing set of 80-20 split.
- Train a Support Vector Machin (SVM) for classification of the digits using the training set.
The following code shows how to train a model using sklearn.

```
clf = svm.SVC()  
clf.fit(X_train, y_train)
```

- Test the model using the test set.
- Experiment with different parameter values for the SVM and see how it performs. Try changing the gamma value to be [0.0001, 0.0005, 0.001, 0.005, 0.01]
- Plot the accuracy value with respect to the change in gamma above.

Q3. Train a k-Nearest Neighbours classifier for handwritten digit recognition with MNIST dataset. Try different parameter settings and study how the performance varies.

- Plot the accuracy vs k while changing the number of neighbours (k) with values [1, 3, 5, 7, 9]

Q4. Train a RandomForest classifier for handwritten digit recognition with MNIST dataset. Try different parameter settings and study how the performance varies.

- Plot the accuracy vs max_depth while changing the max depth parameter with values [1, 2, 4, 8, 16]

Q5. Train a Gaussian Naive Bayes classifier for handwritten digit recognition with the MNIST dataset.

Q6. Do a comparison between the four classifiers (SVM, kNN, RandomForest and Naïve Bayes) by plotting the best performing accuracy value for each classifier in a bar chart.